Least squares

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1 Motivation

So let's start by an example, assuming we have a set of data

x	$x_0 = -1$	$x_1 = 0$	$x_2 = 2$	$x_3 = 4$
y	$y_0 = -1$	$y_1 = 2$	$y_2 = 1$	$y_3 = 3$

Recall that in standard polynomial interpolation, we would need to find a polynomial of degree 3, denoted by $p_3(x)$, such that

$$p_3(x_i) = y_i, \quad i = 0, 1, 2, 3.$$

If the data involve error, this error will be carried over by the interpolating polynomial $p_3(x)$.

Now we ask if we can fit the given data set by using a straight line, a polynomial of degree 1. This means that we need to find a polynomial

$$p_1(x) = a_0 + a_1 x \tag{1}$$

such that

$$p_1(x_i) = y_i, \quad i = 0, 1, 2, 3.$$

Recall that the above problem can be written into a linear system

$$\begin{aligned} a_0 - a_1 &= -1, \\ a_0 &= 2, \\ a_0 + 2a_1 &= 1, \\ a_0 + 4a_1 &= 3. \end{aligned}$$

☹ **There is no solution.**

So what shall we do when there is no solution. The answer is to do the **least squares** approximation.

The least squares method finds a linear function $p_1(x)$ that fits the four points in a best possible way. This procedure can be described as follows.

First, we define the error between the straight line $p_1(x)$ of (1) and the given set of data as follows:

$$\psi(a_0, a_1) = \sum_{i=0}^{3} [p_1(x_i) - y_i]^2.$$

Next, we adjust the parameters a_0 and a_1 to **minimize** the error function $\psi(a_0, a_1)$. To carry out the details for our example, we notice that the error function is really given by

$$\psi(a_0, a_1) = \sum_{i=0}^{3} [a_0 + a_1 x_i - y_i]^2.$$

which can be expressed as

$$\psi(a_0, a_1) = (a_0 - a_1 + 1)^2 + (a_0 - 2)^2 + (a_0 + 2a_1 - 1)^2 + (a_0 + 4a_1 - 3)^2.$$

The function $\psi(a_0, a_1)$ attains its minimum value at a point (a_0, a_1) where the gradient of $\psi(a_0, a_1)$ vanishes. Therefore, we need to solve the following system of linear equations:

$$\begin{aligned}
\frac{\partial \psi(a_0, a_1)}{\partial a_0} &= 2(a_0 - a_1 + 1) + 2(a_0 - 2) + 2(a_0 + 2a_1 - 1) + 2(a_0 + 4a_1 - 3) \\
&= 8a_0 + 10a_1 - 10 = 0, \\
\frac{\partial \psi(a_0, a_1)}{\partial a_1} &= -2(a_0 - a_1 + 1) + 4(a_0 + 2a_1 - 1) + 8(a_0 + 4a_1 - 3) \\
&= 10a_0 + 42a_1 - 30 = 0.
\end{aligned}$$

This system of linear equations can then be rewritten as

$$\begin{aligned}
4a_0 + 5a_1 &= 5, \\
5a_0 + 21a_1 &= 15,
\end{aligned}$$

which can be solved using Gaussian elimination to give

$$a_0 = \frac{30}{59},\ a_1 = \frac{35}{59}.$$

As a result, the straight line that best fits the given data set in the **least square sense** is

$$y = \frac{30}{59} + \frac{35}{59}x.$$

2 Least Squares fitting

It can be seen that the method of least squares involves two main steps:

- Formulate the total error.
- Determine parameters (coefficients) to minimize the total error.

Let's do the same procedure to another example.

Example 2.1. Consider the least squares fit to the data by a straight line: $p_1(x) = a_0 + a_1 x$; that is, the

x	$x_0 = 1$	$x_1 = 3$	$x_2 = 4$	$x_3 = 5$
y	$y_0 = 2$	$y_1 = 4$	$y_2 = 3$	$y_3 = 1$

coefficients a_0 and a_1 are to be determined. We have

$$\begin{aligned}
\psi(a_0, a_1) &= (p_1(1) - 2)^2 + (p_1(3) - 4)^2 + (p_1(4) - 3)^2 + (p_1(5) - 1)^2 \\
&= (a_0 + a_1 - 2)^2 + (a_0 + 3a_1 - 4)^2 + (a_0 + 4a_1 - 3)^2 + (a_0 + 5a_1 - 1)^2.
\end{aligned}$$

Observe that $\psi(a_0, a_1)$ is quadratic in the unknown coefficients a_0 and a_1. For a minimum of $\psi(a_0, a_1)$, the unknowns a_0 and a_1 must satisfy the linear equations

$$\begin{aligned}
\frac{\partial \psi(a_0, a_1)}{\partial a_0} &= 2(a_0 + a_1 - 2) + 2(a_0 + 3a_1 - 4) + 2(a_0 + 4a_1 - 3) + 2(a_0 + 5a_1 - 1) \\
&= -20 + 8a_0 + 26a_1 = 0, \\
\frac{\partial \psi(a_0, a_1)}{\partial a_1} &= 2(a_0 + a_1 - 2) + 6(a_0 + 3a_1 - 4) + 8(a_0 + 4a_1 - 3) + 10(a_0 + 5a_1 - 1) \\
&= -62 + 26a_0 + 102a_1 = 0;
\end{aligned}$$

that is,

$$4a_0 + 13a_1 = 10,$$
$$13a_0 + 51a_1 = 31.$$

Gaussian elimination yields the solution

$$a_0 = \frac{107}{35}, \quad a_1 = -\frac{6}{35}. \tag{2}$$

Therefore, the straight line that best fits the given data set in the least squares sense is

$$p_1(x) = \frac{107}{35} - \frac{6}{35}x.$$

Generally, if we consider fitting the data using a polynomial written in power series form

$$p_m(x) = a_0 + a_1 x + \cdots + a_m x^m, \tag{3}$$

then $\psi(a_0, a_1, \cdots, a_m)$ is quadratic in the unknown coefficients $a_0, a_1, \cdots, a_m$. For the data $\{(x_r, f_r)\}_{r=0}^n$, we have

$$\begin{aligned}\psi(a_0, a_1, \cdots, a_m) &= \sum_{r=0}^{n} (p_m(x_r) - f_r)^2 \\ &= (p_m(x_0) - f_0)^2 + (p_m(x_1) - f_1)^2 + \cdots + (p_m(x_n) - f_n)^2.\end{aligned}$$

The coefficients $a_0, a_1, \cdots, a_m$ are determined by solving the linear system

$$\frac{\partial \psi(a_0, a_1, \cdots, a_m)}{\partial a_0} = 0, \tag{4}$$
$$\frac{\partial \psi(a_0, a_1, \cdots, a_m)}{\partial a_1} = 0, \tag{5}$$
$$\vdots \tag{6}$$
$$\frac{\partial \psi(a_0, a_1, \cdots, a_m)}{\partial a_m} = 0, \tag{7}$$

For each value $j = 0, 1, \cdots, m$, the linear equation

$$\frac{\partial \psi(a_0, a_1, \cdots, a_m)}{\partial a_j} = 0$$

is formed as follows. Observe that

$$\frac{\partial p_m(x_r)}{\partial a_j} = x_r^j.$$

Then, by the chain rule,

$$\begin{aligned}\frac{\partial \psi(a_0, a_1, \cdots, a_m)}{\partial a_j} &= \frac{\partial}{\partial a_j}\left[(p_m(x_0) - f_0)^2 + (p_m(x_1) - f_1)^2 + \cdots + (p_m(x_n) - f_n)^2\right] \\ &= 2\left[(p_m(x_0) - f_0)\frac{\partial p_m(x_0)}{\partial a_j} + (p_m(x_1) - f_1)\frac{\partial p_m(x_1)}{\partial a_j} + \cdots + (p_m(x_n) - f_n)\frac{\partial p_m(x_n)}{\partial a_j}\right] \\ &= 2\left[(p_m(x_0) - f_0)x_0^j + (p_m(x_1) - f_1)x_1^j + \cdots + (p_m(x_n) - f_n)x_n^j\right] \\ &= 2\sum_{r=0}^{n}(p_m(x_r) - f_r)x_r^j \\ &= 2\sum_{r=0}^{n} x_r^j p_m(x_r) - 2\sum_{r=0}^{n} f_r x_r^j.\end{aligned}$$

Substituting the power series form for the polynomial $p_m(x_r)$, (3) ,leads to

$$\begin{aligned}\frac{\partial\psi(a_0,a_1,\cdots,a_m)}{\partial a_j} &= 2\left[\sum_{r=0}^{n} x_r^j(a_0+a_1x_r+\cdots+a_mx_r^m)-\sum_{r=0}^{n} f_rx_r^j\right]\\ &= 2\left[a_0\sum_{r=0}^{n} x_r^j+a_1\sum_{r=0}^{n} x_r^{j+1}+\cdots+a_m\sum_{r=0}^{n} x_r^{j+m}-\sum_{r=0}^{n} f_rx_r^j\right]\end{aligned}$$

Therefore, the equations $\frac{\psi(a_0,a_1,\cdots,a_m)}{\partial a_j}=0, j=0,1,\cdots,m$, may be rewritten as the **normal equations:**

$$a_0\sum_{r=0}^{n} x_r^j+a_1\sum_{r=0}^{n} x_r^{j+1}+\cdots+a_m\sum_{r=0}^{n} x_r^{j+m}=\sum_{r=0}^{n} f_rx_r^j, j=0,1,\cdots,m.$$

In matrix form, the normal equations may be written as

$$\begin{bmatrix} \sum_{r=0}^{n} 1 & \sum_{r=0}^{n} x_r & \cdots & \sum_{r=0}^{n} x_r^m \\ \sum_{r=0}^{n} x_r & \sum_{r=0}^{n} x_r^2 & \cdots & \sum_{r=0}^{n} x_r^{m+1} \\ \vdots & \vdots & & \vdots \\ \sum_{r=0}^{n} x_r^m & \sum_{r=0}^{n} x_r^{m+1} & \cdots & \sum_{r=0}^{n} x_r^{2m} \end{bmatrix}\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix}=\begin{bmatrix} \sum_{r=0}^{n} f_r \\ \sum_{r=0}^{n} f_rx_r \\ \vdots \\ \sum_{r=0}^{n} f_rx_r^m \end{bmatrix}$$

The coefficient matrix of this system is symmetric.

Example 2.2. To compute a straight line fit a_0+a_1x to the data $\{(x_r,f_r)\}_{r=0}^n$, we set $m=1$ in the normal equations to give

$$a_0\sum_{r=0}^{n} 1+a_1\sum_{r=0}^{n} x_r=\sum_{r=0}^{n} f_r, \tag{8}$$

$$a_0\sum_{r=0}^{n} x_r+a_1\sum_{r=0}^{n} x_r^2=\sum_{r=0}^{n} f_rx_r. \tag{9}$$

Substituting the data from Example 1, we have the normal equations

x	$x_0=1$	$x_1=3$	$x_2=4$	$x_3=5$
y	$y_0=2$	$y_1=4$	$y_2=3$	$y_3=1$

$$\begin{aligned}a_0(1+1+1+1)+a_1(1+3+4+5)&=(2+4+3+1),\\ a_0(1+3+4+5)+a_1(1^2+3^2+4^2+5^2)&=(1\times2+3\times4+4\times3+5\times1),\end{aligned}$$

that is,

$$\begin{aligned}4a_0+13a_1&=10,\\ 13a_0+51a_1&=31,\end{aligned}$$

which give the same result as in Example 1.

The next example shows how a quadratic polynomial can be determined in the least squares sense for a given set of data.

Example 2.3. To compute a quadratic fit $a_0+a_1x+a_2x^2$ to the data $\{(x_r,f_r)\}_{r=0}^n$, we set $m=2$ in the normal equations to give

$$a_0\sum_{r=0}^{n} 1+a_1\sum_{r=0}^{n} x_r+a_2\sum_{r=0}^{n} x_r^2=\sum_{r=0}^{n} f_r, \tag{10}$$

$$a_0\sum_{r=0}^{n} x_r+a_1\sum_{r=0}^{n} x_r^2+a_2\sum_{r=0}^{n} x_r^3=\sum_{r=0}^{n} f_rx_r, \tag{11}$$

$$a_0\sum_{r=0}^{n} x_r^2+a_1\sum_{r=0}^{n} x_r^3+a_2\sum_{r=0}^{n} x_r^4=\sum_{r=0}^{n} f_rx_r^2 \tag{12}$$

3 Least Squares from the perspective of linear systems

We want to apply the principle of least squares to solve inconsistent system of the form:

$$\sum_{j=1}^{n} a_{ij}x_j = b_i, \quad i = 1, 2, \cdots, m; \tag{13}$$

that is

$$Ax = b,$$

where

$$A = (a_{ij})_{m\times n},\ x = [x_1, \cdots, x_n]^T,\ b = [b_1, \cdots, b_m]^T.$$

Here, $m > n$; that is, the number of equations is more than the number of unknowns in the system of linear system (13). Generally speaking, this system of linear equations is inconsistent in the sense that there is no vector $x = (x_1, x_2, \cdots, x_n)^T$ which satisfies all the equations in (13). In this case, an alternative is to minimize the sum of the squares of the residuals to obtain the least squares solution of the linear system. This solution can be determined in the following way.

- The sum of the squares of the residual is given by

$$\psi(x_1, x_2, \cdots, x_n) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij}x_j - b_i \right)^2.$$

- To find the minimization point for the error function $\psi(x_1, x_2, \cdots, x_n)$, we take the partial derivative of ψ with respect to each variable x_k, yielding

$$\frac{\partial \psi}{\partial x_k} = 2\sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij}x_j - b_i \right) a_{ik},$$

  for $k = 1, 2, \cdots, n$. Therefore, the minimization point $x = (x_1, \cdots, x_n)$ must be a solution of the linear system

$$\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}a_{ik}x_j = \sum_{i=1}^{m} a_{ik}b_i,\ k = 1, 2, \cdots, n.$$
$$\sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}a_{ik}x_j = \sum_{i=1}^{m} a_{ik}b_i,\ k = 1, 2, \cdots, n.$$

  By setting

$$c_{kj} = \sum_{i=1}^{m} a_{ij}a_{ik},\ d_k = \sum_{i=1}^{m} a_{ik}b_i, \tag{14}$$

  we see that the minimization point $x = (x_1, x_2, \cdots, x_n)$ is the solution of the system of linear equations:

$$\sum_{j=1}^{n} c_{kj}x_j = d_k,\ k = 1, 2, \cdots, n. \tag{15}$$

Example 3.1. To compute the system

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \tag{16}$$

that is,

$$A_{3\times 2}x_{2\times 1} = b_{3\times 1}. \tag{17}$$

From (14), we have

$$\left[\begin{array}{ll} c_{11} = \sum_{i=1}^{3} a_{i1}a_{i1} & c_{12} = \sum_{i=1}^{3} a_{i2}a_{i1} \\ c_{21} = \sum_{i=1}^{3} a_{i1}a_{i2} & c_{22} = \sum_{i=1}^{3} a_{i2}a_{i2} \end{array} \right], \left[\begin{array}{l} d_1 = \sum_{i=1}^{3} a_{i1}b_i \\ d_2 = \sum_{i=1}^{3} a_{i2}b_i \end{array} \right] \tag{18}$$

and

$$\begin{aligned} c_{11}x_1 + c_{12}x_2 &= d_1, \\ c_{21}x_1 + c_{22}x_2 &= d_2. \end{aligned}$$

☺ For Example 3.1, we find that

$$A^T A = \left[\begin{array}{lll} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{array} \right] \left[\begin{array}{ll} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{array} \right] = \left[\begin{array}{ll} c_{11} = \sum_{i=1}^{3} a_{i1}a_{i1} & c_{12} = \sum_{i=1}^{3} a_{i2}a_{i1} \\ c_{21} = \sum_{i=1}^{3} a_{i1}a_{i2} & c_{22} = \sum_{i=1}^{3} a_{i2}a_{i2} \end{array} \right] = C$$

and

$$A^T b = \left[\begin{array}{lll} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{array} \right] \left[\begin{array}{l} b_1 \\ b_2 \\ b_3 \end{array} \right] = \left[\begin{array}{l} d_1 = \sum_{i=1}^{3} a_{i1}b_i \\ d_2 = \sum_{i=1}^{3} a_{i2}b_i \end{array} \right] = d$$

Therefore, the least squares solution of the matrix problem

$$Ax = b$$

is given by the solution of the normal equations

$$A^T Ax = A^T b.$$

Linear Algebra: geometric solution, orthogonality; refer to Chapter 5 and Section 7.7 in [1] The vector u and v are orthogonal, if $u^T \cdot v = 0$.

$$(b - A\bar{x}) \perp \{Ax | x \in R^n\} \tag{19}$$

Then

$$(Ax)^T (b - A\bar{x}) = 0 \text{ for all } x \text{in } R^n. \tag{20}$$

Then

$$x^T A^T (b - A\bar{x}) = 0 \text{ for all } x \text{in } R^n, \tag{21}$$

meaning that the n-dimensional vector $A^T(b - A\bar{x})$ is orthogonal to every vector x in R^n. There is only one way for that to happen:

$$A^T (b - A\bar{x}) = 0, \tag{22}$$

where leads to the normal equations

$$A^T A\bar{x} = A^T b. \tag{23}$$

The following summarizes the procedure in the least squares approach for inconsistent linear system of the form $Ax = b$:

Step 1. Find the transpose of A, denoted by A^T.

Step 2. Multiply the system $Ax = b$ by A^T: $A^T Ax = A^T b$.

Step 3. Find the solution for the normal system: $A^T Ax = A^T b$.

So let's to do an example.

Example 3.2. Find the least squares solution of the inconsistent system

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \tag{24}$$

Solution (by hand, purpose for check the code!):

Step 1. Find the transpose of A,

$$A^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \end{bmatrix} \tag{25}$$

Step 2. Multiply the system $Ax = b$ by A^T: $A^TAx = A^Tb$, where

$$A^TA = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix},\ A^Tb = \begin{bmatrix} 2 \\ -3 \end{bmatrix}. \tag{26}$$

Step 3. Find the solution for the normal system: $A^TAx = A^Tb$, that is,

$$\begin{aligned} 3x_1 + 2x_2 &= 2, \\ 2x_1 + 6x_2 &= -3. \end{aligned}$$

The solution is given by

$$x_1 = \frac{9}{7},\ x_2 = -\frac{13}{14}.$$

Solve it by Matlab Commands:

```
>> A = [1 1 1; 1 -1 2]' %   the matrix A
>> y = [1 2 -1]'
>> c=A'*A \ A'*y % Solving the normal equations: (A^t A) c= A^t y
% gives c=[9/7 -13/14]'
```

Example 3.3. Back to the Example in the least squares fitting,
See the white board illusion!

Solve it by Matlab Commands:

```
>> x=[-1 0 2 4] % HAS TO BE a COLUMN vector
>> y=[-1 2 1 3] % Also a COLUMN vector
>> A=[x x.^0] % a fast way to build the matrix A
>> c=A'*A \ A'*y % Solving the normal equations: (A^t A) c= A^t y
% gives c=[35/59 30/59]'
```

The least squares line that fits the data is $y = \frac{35}{59}x + \frac{30}{59}$.

Plot the data and the least squares line:

```
>> xh=-1:0.02:4; % many grid points for [-1, 4]
>> yh=polyval(c,xh);%evaluating y=35/59x+30/59 @ xh
>> plot(x,y,ob,xh,yh,m)
```

4 Conditioning of linear system

The condition number is the relative change in the solution divided by a relative change in the data. In this case the data are A and b, and the solution is x.

For simplicity we start by allowing perturbations to b only while A remains unchanged. Let $Ax = b$ be perturbed to

$$A(x + h) = b + d.$$

We seek to bound $\|h\|$ in terms of $\|d\|$:

$$\begin{aligned} Ax + Ah &= b + d, \\ Ah &= d, \\ h &= A^{-1}d, \\ \|h\| &\leq \|A^{-1}\|\|d\|, \end{aligned}$$

where $\|\cdot\|$ can be a norm of

$$\|A\|_\infty = \max_{1\leq i\leq n} \sum_{j=1}^{n} |A_{ij}|. \tag{27}$$

Since furthermore $b = Ax$ and therefore $\|b\| \leq \|A\|\|x\|$, we derive

$$\frac{\frac{\|h\|}{\|x\|}}{\frac{\|d\|}{\|b\|}} = \frac{\|h\|\|b\|}{\|d\|\|x\|} \leq \frac{(\|A^{-1}\|\|d\|)(\|A\|\|x\|)}{\|d\|\|x\|} = \|A^{-1}\|\|A\|. \tag{28}$$

It is possible to show that this bound is "tight", in the sense that the inequalities are in fact equalities for some choices of b and d. We define the matrix condition number

$$\kappa(A) = \|A^{-1}\|\|A\|. \tag{29}$$

One observation:

$$1 = \|I\| = \|AA^{-1}\| \leq \|A\|\|A^{-1}\| = \kappa(A). \tag{30}$$

4.1 Conditioning of least squares

We have seen that the least squares problem reduced to solving the normal equations $A^TA\bar{x} = A^Tb$. How accurately can the least squares solution $\bar{x}$ be determined? We carry out a double precision numerical experiment to test this question, by solving the normal equations in a case where the correct solution is known.

Example 4.1. Let $x_1 = 2.0, x_2 = 2.2, x_3 = 2.4, \cdots, x_{11} = 4.0$ be equally spaced points in $[2, 4]$, and set $y_i = 1 + x_i + x_i^2 + x_i^3 + x_i^4 + x_i^5 + x_i^6 + x_i^7$ for $1 \leq i \leq 11$. Use the normal equations to find the least square polynomial $P(x) = c_1 + c_2x + \cdots + c_8x^7$ fitting the (x_i, y_i).

A degree 7 polynomial is being fit to 11 data points lying on the degree 7 polynomial $P(x) = 1+x+x^2+x^3+x^4+x^5+x^6+x^7$. Obviously, the correct least squares solution is $c_1 = c_2 = \cdots = c_8 = 1$. Substituting the data points into the model $P(x)$ yields the system $Ac = b$:

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^7 \\ 1 & x_2 & x_2^2 & \cdots & x_2^7 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{11} & x_{11}^2 & \cdots & x_{11}^7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_8 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_1 1 \end{bmatrix}. \tag{31}$$

We use MATLAB to solve the normal equations:

```
>> x = (2+(0:10)/5)';
>> y = 1 + x + x.^2 + x.^3 + x.^4 + x.^6 + x.^7;
>> A = [x.^0 x x.^2 x.^3 x.^4 x.^5 x.^6 x.^7];
>> c = (A'*A)\(A'*y)

c=
     1.5134
    -0.2644
     2.3211
     0.2408
     1.2592
     0.9474
     1.0059
     0.9997

>> cond(A'*A)

ans=
  1.4359e+019
```

Solving the normal equations in double precision cannot deliver an accurate value for the least squares solution. The condition number of A^TA is too large to deal with in double precision arithmetic, and the normal equations are ill-conditioned, even though the original least squares problem is moderately conditioned. There is clearly room for improvement in the normal equations approach to least squares; there are fours ways summarized in [3], which can be listed in the following.

1 $A^TA\hat{x} = A^Tb$ can be solved directly when A has independent columns. It is fine for small problems!!!
2 The Gram-Schmidt idea produces orthogonal column in Q. Then $A = QR$; also refer to Section 4.3 in [2].
3 The SVD of A leads to its pseudoinverse A^+. Then $\hat{x} = A^+b$: One short formula;
also refer to Chapter 12 in [2].
4 Minimize $\|b - Ax\|^2 + \delta^2\|x\|^2$. The penalty changes the normal equations to $(A^TA + \delta^2 I)x_\delta = A^Tb$.
Now the matrix is invertible and x_δ goes to $\hat{x}$ as $\delta \to 0$.

Example 4.2. Use the full QR factorization to solve the least squares problem of Example 4.1.

The normal equations were notably unsuccessful in solving this least squares problem of 11 equations in 8 variable. We use the MATLAB command `qr` to carry out an alternative approach:

```
>> x=(2+(0:10)/5)';
>> y=1+x+x.^2+x.^3+x.^4+x.^5+x.^6+x.^7;
>> A=[x.^0 x x.^2 x.^3 x.^4 x.^5 x.^6 x.^7];
>> [Q,R]=qr(A);
>> b=Q'*y;
>> c=R(1:8,1:8)\b(1:8)
c=
0.99999991014308
1.00000021004107
0.99999979186557
1.00000011342980
0.99999996325039
1.00000000708455
0.99999999924685
1.00000000003409
```

Six decimal places of the correct solution $c = [1, \cdots, 1]$ are found by using QR factorization.

4.2 MATLAB commands polyfit and polyval

The MATLAB commands `polyfit` and `polyval` are designed not only to interpolate data, but also to fit data with polynomial models. For n input data points, `polyfit` used with input degree $n-1$ returns the coefficients of the interpolating polynomial of degree $n-1$. If the input degree is less than $n-1$, `polyfit` will instead find the best least squares polynomial of that degree. For exmaple,

```
>> x0 = [-1  0  1  2]';
>> y0 = [1   0  0  -2]';
>> c = polyfit(x0,y0,2);
>> x = -1:0.02:2;
>> y = polyval(c,x);
>> plot(x0,y0,'o',x,y)
```

References

[1] S. J. Leon. *Linear algebra with applications.*

[2] T. Sauer. Numerical Analysis, 2019.

[3] G. Strang. *Linear algebra and learning from data*, volume 4. Wellesley-Cambridge Press Cambridge, 2019.